

Preferendus-IMAP-IGS

Preference-Based Optimisation — IMAP Inter-Generational Solver (IGS) Framework

About

This is the home of the IMAP Inter-Generational Solver (IGS) framework - a collection of preference-based optimisation solvers built on the IMAP (Integrative Maximisation of Aggregated Preferences) aggregator. The page hosts four repositories, each a distinct IGS instantiation targeting different problem classes: IGS-GA-I, IGS-GA-II, IGS-BRKGA, and IGS-SI.

The open-source Preferendus software for preference-based optimisation used for the IMAP demonstrators in the ODESYS book [1] is publicly available via the GitHub repository: <https://github.com/TUdelft-Odesys>, including the early IMAP/IGS 1.0 implementation.

Repositories

IGS-GA-I

<https://github.com/TUdelft-IMAP-IGS/IGS-GA-I>

The original IGS-GA algorithm — the foundational Inter-Generational Solver using a Genetic Algorithm backbone. This is a user-friendly implementation, equivalent to the original version linked to all ODESYS book examples. It integrates IMAP preference aggregation directly into standard GA survival selection.

IGS-GA-II

<https://github.com/TUdelft-IMAP-IGS/IGS-GA-II>

An improvement on IGS-GA-I, built on the NSGA-II genetic loop (crossover, mutation, mating) but replacing non-dominated sorting and crowding distance with IMAP affine aggregation survival. Designed for benchmarking against reference test suites (e.g. DAS-CMOP). Supports multi-stakeholder preference models, constraint handling via the Constraint Dominance Principle (CDP), tournament and truncation selection, and an optional cross-generation archive to stabilise IMAP Z-score normalisation.

IGS-BRKGA

<https://github.com/TUdelft-IMAP-IGS/IGS-BRKGA>

A Biased Random-Key Genetic Algorithm (BRKGA) instantiation of the IMAP-IGS framework, developed for combinatorial and constraint-heavy optimisation problems. Used as the solver for the Heterogeneous Vessel Allocation and Scheduling Problem (HVASP), and the Multi-Objective Vehicle Routing Problem with Time Windows (MO-VRPTW).

IGS-SI

<https://github.com/TUdelft-IMAP-IGS/IGS-SI>

A Swarm Intelligence (SI) instantiation of the IMAP-IGS framework, targeting graph-based and routing optimisation problems. Implements the Preferendus method via Ant Colony Optimisation (ACO), with the same preference-aggregation interface as the GA-based solvers. Applicable to directed graph problems such as route planning and Markov-chain modelling.

IMAP-IGS Terminology

IGS-GA-II: A GA which inherits the genetic loop (crossover, mutation, mating) from NSGA-II but replaces the non-dominated sorting and crowding distance with IMAP affine aggregation survival.

IMAP (Integrative Maximisation of Aggregated Preferences):

The preference function $P_{k,i}$ maps an actor-independent performance function vector F to a corresponding preference score, reflecting the goal-oriented evaluation of actor k , such that $P_{k,i}(F) \in [0, 100]$, where $P_{k,i}$ acts component-wise on the i^{th} element of F . The relatively worst (least preferred) and best (most preferred) performance levels are assigned preference scores of 0 and 100, respectively, representing an arbitrary reference scaling since preferences possess only relative meaning within a one-dimensional affine space, i.e., all preference values are defined only up to relative differences and not in absolute terms [Wolfert (2026)].

Survival Selection

IMAPSurvival: The two-pass survival operator that replaces NSGA-II's RankAndCrowdingSurvival.

Two-pass aggregation:

- First pass: runs IMAP over the full pool (parents + offspring) to identify and discard low-quality candidates (score ≤ 40).
- Second pass: runs IMAP over the surviving candidates only, producing the final scores used for selection.

Discard phase / threshold (score ≤ 40): After the first pass, any individual scoring ≤ 40 is removed from the candidate pool before the second pass. Skipped if it would leave fewer than N survivors. Under Constraint dominance principle (CDP), feasible individuals are always protected from discard when there are fewer feasible solutions than N .

Candidates: The pool that remains after the discard phase. The second-pass IMAP scores are computed on this set.

Truncation selection: Deterministically keeps the top N candidates ranked by final IMAP score. Under CDP the sort is lexicographic: feasible individuals rank above infeasible ones, then by IMAP score within each group.

Tournament selection: Runs N binary tournaments on the candidate population. Each tournament samples two candidates at random. The higher-scoring (and feasibility-preferred under CDP) candidate wins. Preserves more genetic diversity than truncation at the cost of weaker selection pressure.

Performance - Preference Model

Stakeholder: A decision-maker whose preferences are encoded as a set of preference functions over the objectives, plus a per-objective weight vector and an overall importance weight.

Stakeholder weights: Relative importance of each stakeholder. Normalised internally so they sum to 1.

Preference function: Function over an objective, mapping that objective to a preference score, reflecting the preference of a stakeholder.

Objective weights: Per-stakeholder importance of each objective. Should sum to 1 per stakeholder.

Performance function: Here, $F = (f_1, \dots, f_I)$ denotes the vector of system performance functions, where each $f_i(x, y, t)$ denotes the i^{th} performance function, and I is the total number of performance functions. These functions describe the physical behaviour of the system, i.e. its capabilities, representing what the system *can* perform. The vector x represents the 'controllable' design-decision vector, comprising both product and process variables that can be deliberately selected and adjusted by the decision maker. The vector y is the 'uncontrollable' parameter vector, containing contextual variables, including external product and process conditions that influence system behaviour but cannot be controlled.

A-fine-aggregator

A-fine-aggregator: Core aggregation function. Steps:

1. Transpose p so rows = individuals, columns = criteria.
2. Z-score normalise each column: $z = \frac{x - \mu}{\sigma}$. If $\sigma = 0$, use $\sigma = 10^{-6}$ to avoid division by zero.
3. Compute $P^* = \sum w_i \cdot z_i$ (weighted centroid over all criteria), where $\sum w_i = 1$.
4. Min-max scale to $[0, 100]$. If all P^* values are identical, return 50 for all.

The endpoints 0 and 100 thus represent the relative minimum and maximum within the alternative set, with the zero point being purely relational rather than an absolute preference origin.

Z-score normalisation: Standardises each criterion to zero mean and unit variance across the pool. This ensures criteria with very different numerical ranges contribute equally to P^* .

P^* (representative preference score): The weighted sum of Z-score-normalised preference values for one individual. The raw output of the aggregation before min-max scaling. See Wolfert [4] for more information.

Constraint Handling

Constraint Dominance Principle (CDP): A feasible individual always beats an infeasible one in any comparison (survival or tournament), regardless of IMAP score. Applies whenever there are constraints.

Constraint Violation (CV): Total constraint violation of an individual: $CV = \sum \max(0, g_j)$ over all constraints. A solution is feasible if and only if $CV = 0$.

CV preference function: An explicit preference applied to each individual's total CV. When set, all stakeholder weights are scaled by $(1 - cv_weight)$ and the CV criterion receives weight cv_weight . When the population is fully feasible, all CV scores ≈ 100 and $std \approx 0$, so the CV term contributes essentially nothing.

CV weight: Fraction of total aggregation weight assigned to the CV criterion. Must be in $(0, 1)$ when cv_pref_fn is set.

Elitism and Best Tracking

Current best: The highest-scoring feasible individual seen across all completed survival passes. Injected into the pool at every generation to update its preference score with the new population. Updated only when a strictly better, genuinely different solution is found in the same generation's second-pass scoring context.

Current_best_gen: The generation index at which `current_best` was last updated.

Hall of fame: Ordered list of all outgoing `current_best` values. Every time `current_best` is replaced, the old one is appended here. Provides a history of all best-found solutions across the run. Note: general concept in GA's.

Elitism guarantee: Before each generation's selection call, `current_best` is injected into the population if not already present. After selection, `current_best` is force-kept in the survivors if it would otherwise be dropped.

Archive

Best archive (memory): Optional storing of previously best individuals. When enabled, archive members are appended to the candidate pool's objective-value dict before the second-pass IMAP scoring. This extends the normalisation reference frame across generations, stabilising Z-score computation.

Archive stabilisation: Archive members participate in Z-score normalisation only; they are never returned as survivors. Including them prevents the score distribution from collapsing when the live candidate pool becomes too homogeneous.

Archive max size: Maximum archive size. When exceeded, we score all archive members against each other and discards the lower-scoring 50%, retaining at least one individual.

Inherited NSGA-II Infrastructure

Population size: Number of individuals maintained in the population across generations.

Population: Container for a set of Individual objects. Carries evaluated objective values (F), constraint values (G), decision variables (X), and any custom attributes (here: `imap_score`, `is_feasible`).

Individual: Single solution with fields X (decision variables), F (objectives), G (constraints), and arbitrary named attributes.

Crossover / Mutation: Standard pymoo operators inherited from NSGA2/GeneticAlgorithm; IMAP-GA does not modify them.

References

[1] Wolfert, A.R.M. et al. (2023). *Open Design Systems (ODESYS)*. TU Delft. Repository: <https://github.com/TUdelft-Odesys>

[2] Özer, E. and Wolfert, A.R.M. (2026). Ordering Is Not Deciding: Towards Preference-Consistent Design--Decision Optimisation. *To be published*.

[3] Timp, L.J.K., Yorke-Smith, N. & Wolfert, A.R.M (2026). Solving highly constrained multi-objective decision problems with preference-guided metaheuristics: The IMAP-IGS framework. *To be published*.

[4] Wolfert, A.R.M. (2026). Unique preference aggregation in design and decision making. *arXiv*. <https://doi.org/10.48550/arXiv.2601.19759>

[5] Wolfert, A.R.M. (2026). *Preference-Based Optimisation for Integrated Design and Group Decision-Making*. arXiv preprint arXiv:2603.19050. Available at: <https://arxiv.org/abs/2603.19050>.
Preferendus software: IMAP-IGS